EXPLORATORY DATA ANALYSIS

CHAPTER 12

Multivariate Visualization

Visualizing several variables together while preserving clarity and analytical meaning

Focus: visual encodings, faceting, heatmaps, parallel coordinates, bubble plots, pairwise matrices, and the challenges of high-dimensional displays.

Technical convention: Matplotlib object-oriented API; Seaborn axes-level functions use explicit Axes objects. This chapter is about visualization, not dimensionality-reduction algorithms.

Learning Objectives

- Explain what makes a visualization multivariate and distinguish it from a collection of unrelated univariate plots.
- Map variables to position, color, size, shape, facets, and ordering without overloading a chart.
- Use faceting and small multiples to compare conditional patterns.
- Construct and interpret heatmaps, parallel-coordinate plots, and bubble plots.
- Use pair plots and correlation heatmaps selectively and understand what they omit.
- Recognize perceptual limitations, scale effects, and clutter in multivariate displays.
- Design a coherent visual analysis that supports exploration without implying causation.

Review of Earlier Chapters

Chapter 1 — What Is EDA?

EDA is iterative investigation. Visualizations should answer analytical questions, support quantitative evidence, and distinguish observations from explanations.

Chapter 2 — Understanding the Dataset

Know the observational unit, variable definitions, units, measurement scales, and granularity. Combining variables from incompatible levels can create misleading displays.

Chapter 3 — Data Quality Assessment

Missingness, invalid values, inconsistent units, duplicates, and data-collection artifacts can distort multivariate patterns. Missingness may vary jointly across several variables.

Chapter 4 — Exploring Numerical Variables

Univariate center, spread, skewness, and outliers influence the scales and visible structure of multivariate plots.

Chapter 5 — Exploring Categorical Variables

Category cardinality and rare levels affect color legends, facets, and grouped comparisons. Combining rare levels requires a defensible domain reason.

Chapter 6 — Exploring Date and Time Variables

Calendar components, elapsed time, and temporal coverage can act as additional dimensions. Trends over time can induce apparent relationships among variables.

Chapter 7 — Descriptive Statistics

Robust summaries and quantiles provide numerical context for multivariate visual comparisons.

Chapter 8 — Understanding Distributions

Marginal distributions remain important even when several variables are shown together. A multivariate display should not conceal the shape of individual variables.

Chapter 9 — Distribution Shape

Skewness, kurtosis, and quantiles help identify variables whose ranges and tails can dominate visual encodings.

Chapter 10 — Visualizing Distributions

Histograms, KDEs, ECDFs, boxplots, violins, and small multiples describe individual distributions and group comparisons.

Chapter 11 — Visualizing Relationships

Scatterplots, hexbin plots, joint plots, pair plots, and categorical comparisons reveal pairwise relationships, overplotting, nonlinear patterns, and subgroup effects.

Review of Chapter 11 Assignment and Project

No assignment submission was included, so this is an expected-reasoning guide rather than grading.

- A scatterplot should be read for direction, form, strength, clusters, changing spread, and unusual points.
- Transparency and aggregation help with overplotting; jitter is a display device and should not be confused with observed coordinates.
- Hexbin and 2D histograms show local counts; bivariate KDE shows a smoothed estimate that depends on bandwidth.
- Categorical–numerical comparisons should show distributions, not only group means.

- Row-normalized crosstabs answer within-row questions; column normalization answers a different question.
- Pooled and subgroup patterns may differ because of group composition or confounding; association does not establish causation.

Expected project learning: select visual forms for variable types and sample size, report pairwise sample sizes, and explain limitations.

Preview of This Chapter

Multivariate visualization tries to show more than two variables in a single analytical view. The central challenge is not how many variables can be encoded, but how many can be interpreted accurately by a human reader. We will use visual channels deliberately and favor several clear coordinated views over one overloaded chart.

1. What Counts as Multivariate Visualization?

A multivariate display represents three or more variables in a coordinated visual structure. Examples include a scatterplot with color for group and size for a fourth variable, a faceted grid, a heatmap, or parallel coordinates.

A chart may encode many variables but still fail analytically. Every added channel increases cognitive load and can create ambiguity. Start with the question and decide which variables are essential.

2. Visual Encodings and Perceptual Strength

Visual channel	Typical use	Caution
Position	Numerical values on x/y axes	Usually among the most accurately judged channels.
Length	Bars or intervals	Use a common baseline for direct magnitude comparisons.
Color hue	Nominal categories	Many hues are difficult to distinguish.
Color lightness	Ordered magnitude	Use perceptually ordered scales; avoid misleading gradients.
Size / area	A third numerical variable	Area is harder to judge than position; scale area, not radius, to value.
Shape	A small number of categories	Too many marker shapes become hard to distinguish.
Faceting	Condition on another variable	Many panels reduce space and attention.
Ordering	Sequence or rank	The ordering rule must be explicit.

A useful design principle: encode the most important quantitative variables using position, and use color or facets for secondary grouping. Avoid assigning many variables to weak or ambiguous channels.

3. Extending a Scatterplot With More Variables

A scatterplot can encode x and y as position, a category as color, and another numerical variable as point size. This is often called a bubble plot.

```
fig, ax = plt.subplots(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x="income",
    y="spending",
    hue="segment",
    size="household_size",
    sizes=(25, 250),
    alpha=0.65,
    ax=ax
)
ax.set(
    title="Spending and income by segment",
    xlabel="Income",
    ylabel="Spending"
)
ax.legend(
    bbox_to_anchor=(1.02, 1),
    loc="upper left"
)
fig.tight_layout()
```

Bubble size should represent area proportionally to the encoded quantity. Seaborn manages marker sizing, but the mapping and legend still need inspection. Large bubbles can hide small ones; transparency and careful ordering may help.

- Use a small number of meaningful categories for hue.
- Check that size values are nonnegative and that the size range is not exaggerated.
- Do not use bubble size for a variable whose exact value must be read precisely.
- Consider separate panels or direct labels if the plot becomes crowded.

4. Faceting and Small Multiples

Faceting repeats a plot for different values of a categorical variable. It is one of the clearest ways to compare conditional relationships while preserving the same visual grammar.

```
g = sns.relplot(
    data=df,
    x="income",
    y="spending",
    col="region",
    hue="segment",
    kind="scatter",
    alpha=0.6,
    col_wrap=3,
    height=3
)
g.set_axis_labels("Income", "Spending")
```

Small multiples work best when panels share scales for direct comparisons. Free scales may reveal within-panel patterns but make between-panel magnitude comparisons harder. State which choice you make.

Facet by rows and columns

```
g = sns.relplot(
    data=df,
    x="income",
    y="spending",
    row="region",
    col="customer_type",
    kind="scatter",
    height=2.6,
    facet_kws={"sharex": True, "sharey": True}
)
g.set_axis_labels("Income", "Spending")
```

A two-way facet grid can become sparse or enormous. Use it only when each panel has enough observations and the combination of groups has a clear interpretation.

5. Heatmaps for Multivariate Tables

A heatmap uses color to encode values in a matrix. Rows and columns often represent categories, variables, or ordered bins. Heatmaps are especially useful when a table is too large to inspect cell by cell.

```
summary = pd.pivot_table(
    df,
    index="region",
    columns="product_type",
    values="revenue",
    aggfunc="median"
)

fig, ax = plt.subplots(figsize=(8, 5))
sns.heatmap(
    summary,
    annot=True,
    fmt=".1f",
    cmap="viridis",
    ax=ax
)
ax.set(
    title="Median revenue by region and product",
    xlabel="Product type",
    ylabel="Region"
)
fig.tight_layout()
```

The aggregation function defines what each cell means. Median revenue, mean revenue, count, and proportion answer different questions. Include units and explain how empty combinations are represented.

Normalize deliberately

Row-normalized heatmaps compare composition within each row; column-normalized heatmaps compare composition within each column. A global normalization compares absolute values. Always state the denominator.

```
composition = pd.crosstab(
    df["region"],
    df["product_type"],
    normalize="index"
)

fig, ax = plt.subplots(figsize=(8, 5))
sns.heatmap(
    composition,
    annot=True,
    fmt=".0%",
    cmap="Blues",
    vmin=0,
    vmax=1,
    ax=ax
)
ax.set(
    title="Product mix within each region",
    xlabel="Product type",
    ylabel="Region"
)
fig.tight_layout()
```

Do not compare cell colors across differently normalized heatmaps without explaining the change in scale.

6. Correlation Heatmaps

A correlation matrix summarizes pairwise linear correlations (or another explicitly selected association measure) among numerical variables. A heatmap can reveal broad clusters of positive and negative pairwise associations.

```
numeric = df.select_dtypes(include="number")
corr = numeric.corr(method="pearson")

fig, ax = plt.subplots(figsize=(8, 6))
sns.heatmap(
    corr,
    cmap="vlag",
    center=0,
    vmin=-1,
    vmax=1,
    annot=True,
    fmt=".2f",
    square=True,
    ax=ax
)
ax.set_title("Pearson correlation matrix")
fig.tight_layout()
```

- Correlation is pairwise: it does not show all higher-order relationships.
- Pearson correlation measures linear association and can miss nonlinear dependence.
- Correlations may be affected by outliers, subgroup mixtures, and restricted ranges.
- Missing-value handling can change the observations used for each pair; pairwise-complete matrices may use different n for different cells.
- Correlation does not establish causation or independence.

Chapter 13 will examine association measures in more depth. Here, the heatmap is a visual overview—not a complete statistical analysis.

7. Reordering Heatmaps and Clustering

Rows and columns can be reordered to place similar profiles near one another. A clustered heatmap uses a similarity or distance measure to group rows or columns. The ordering can reveal structure, but it depends on the chosen variables, scaling, distance, and linkage method.

```
g = sns.clustermap(
    corr,
    cmap="vlag",
    center=0,
    vmin=-1,
    vmax=1,
    figsize=(8, 8)
)
```

Clustering is an exploratory ordering device here, not proof that natural groups exist. Correlation-based distances, Euclidean distances, and scaled variables can produce different arrangements. Avoid interpreting dendrogram branches as definitive categories.

8. Parallel Coordinates

Parallel coordinates place each numerical variable on a separate vertical axis. Each observation becomes a line crossing all axes at its values. This can reveal profiles, clusters, and unusual combinations across several variables.

```
from pandas.plotting import parallel_coordinates

plot_df = df[
    ["group", "age", "income", "spending", "visits"]
].dropna()

fig, ax = plt.subplots(figsize=(10, 5))
parallel_coordinates(
    plot_df,
    class_column="group",
    cols=["age", "income", "spending", "visits"],
    ax=ax,
    alpha=0.35
)
ax.set_title("Parallel coordinates by group")
ax.set_ylabel("Original variable scale")
fig.tight_layout()
```

Parallel coordinates are sensitive to variable scales. A variable with a much larger numeric range can dominate the visual geometry. Standardizing or min–max scaling may help compare profiles, but changes the meaning of the axes; clearly label normalized values and retain the original summaries.

- Use a manageable number of variables and observations.
- Order axes intentionally; adjacent axes determine which relationships are easiest to see.
- Many crossing lines create clutter; use transparency, sampling, or group summaries.
- Do not interpret line crossings as causal interactions.

9. Bubble and Dot Matrix Displays

Bubble charts encode a value using marker area at positions defined by two other variables. They can be useful for a small matrix of categories or a limited number of observations.

```
fig, ax = plt.subplots(figsize=(8, 5))
scatter = ax.scatter(
    df["x_metric"],
    df["y_metric"],
    s=df["volume"] * 4,
    c=df["margin"],
    cmap="viridis",
    alpha=0.65
)
fig.colorbar(scatter, ax=ax, label="Margin")
ax.set(
    title="Performance, volume, and margin",
    xlabel="Metric X",
    ylabel="Metric Y"
)
fig.tight_layout()
```

This example's size scaling is illustrative. In a professional chart, map area to value, inspect the size range, handle zero values, and provide a clear size legend. If size and color both encode numerical values, ensure readers can decode each independently.

10. Choosing Among Multivariate Displays

Analytical need	Good starting display	Main limitation
Compare relationship across groups	Faceted scatterplots	Many panels can become sparse.
Summarize many pairwise correlations	Correlation heatmap	Pairwise and often linear only.
Compare category combinations	Aggregated heatmap	Cell meaning depends on aggregation/normalization.
Inspect profiles across several metrics	Parallel coordinates	Scale and line clutter.
Show 3–4 variables in a small dataset	Bubble scatterplot	Size is difficult to judge precisely.
Scan several numerical variables	Pair plot	Quadratic growth in panels and clutter.
Compare subgroup distributions	Faceted histograms/boxplots	Requires careful shared-scale decisions.

11. Scaling, Units, and Transformations in Displays

Multivariate displays often combine variables measured in different units. A parallel-coordinate plot, heatmap, or distance-based ordering may be dominated by large-scale variables.

- Preserve original units when they are central to interpretation.
- Use standardized or normalized display values only when the purpose is profile comparison, and label the transformation.
- Do not silently mix raw dollars, percentages, counts, and years on one numeric axis.
- Logarithmic axes can reveal structure across orders of magnitude but change visual distances; label them clearly and handle zeros explicitly.
- Scaling a visualization does not change the underlying data unless you deliberately transform the analysis dataset.

12. Missingness in Multivariate Displays

Rows missing any of several plotted variables may be excluded, reducing the displayed sample. A complete-case pair plot can use a smaller and potentially selected subset.

```
columns = ["age", "income", "spending", "visits"]
view = df[columns].copy()

missing_by_column = view.isna().sum()
complete_rows = view.dropna()

print("Original rows:", len(view))
print("Complete rows:", len(complete_rows))
print(missing_by_column)
```

Report how many rows remain and whether missingness differs by group. Do not assume complete cases represent all observations.

13. A Practical Multivariate Visualization Workflow

- Start with the analytical question and observational unit.
- Choose a small set of variables with clear definitions and units.
- Inspect their individual distributions and data quality.
- Identify variable types and meaningful groupings.
- Choose a primary visual encoding; reserve secondary channels for supporting information.

- Check sample sizes, missingness, scales, and overplotting.
- Create a second complementary view to test whether the pattern is robust to the display choice.
- Summarize visible patterns, uncertainty, and limitations without causal claims.

14. Common Mistakes

- Encoding too many variables in one chart.
- Using color palettes with too many indistinguishable categories.
- Mapping a numerical value to marker radius rather than area.
- Using different facet scales while implying direct magnitude comparison.
- Treating correlation heatmaps as complete representations of dependence.
- Interpreting clustered heatmap ordering as proof of natural groups.
- Using parallel coordinates without considering scale and axis order.
- Failing to state the aggregation function or normalization denominator in heatmaps.
- Ignoring missing rows and changing sample sizes.
- Presenting normalized values without saying what normalization means.
- Using 3D plots when a 2D faceted view would be easier to interpret.

Chapter Summary

- Multivariate displays encode three or more variables, but more encodings do not automatically mean more insight.
- Position is generally a strong channel for quantitative values; color, size, shape, and facets should be used deliberately.
- Bubble plots can add dimensions but marker area is difficult to judge precisely.
- Faceting supports conditional comparisons, especially with shared scales.
- Heatmaps summarize matrices; aggregation and normalization determine what colors mean.
- Correlation heatmaps are pairwise summaries and can miss nonlinear or higher-order structure.
- Parallel coordinates show profiles but are sensitive to scale, axis order, and clutter.
- Missingness, sample size, and units remain essential to interpretation.

Assignment

Part A — Conceptual Questions

- What makes a visualization multivariate?
- Why is position generally preferable to color or size for encoding a key numerical variable?
- Explain why marker area should be proportional to the quantity in a bubble plot.
- When are small multiples preferable to one plot with many colors?
- What does a heatmap cell represent, and why must the aggregation function be stated?
- Compare row-normalized and column-normalized heatmaps.
- What can a correlation heatmap reveal, and what can it not establish?
- Why do scaling and axis order matter in parallel coordinates?
- Why can a complete-case multivariate plot be misleading?
- Describe two ways to reduce visual clutter without hiding important patterns.

Part B — Build Three Multivariate Views

Use a dataset with at least four useful variables. Create:

- A scatterplot with x/y position, a categorical hue, and one numerical size variable.
- A faceted scatterplot or small-multiple distribution comparison using a meaningful group.
- A heatmap summarizing a category-by-category or variable-by-variable table.

For each, state the question, variable meanings, sample size, missing-value handling, encodings, and the main limitation.

Part C — Parallel Coordinates or Pair Plot

Choose a manageable set of numerical variables. Create either a parallel-coordinate plot or a pair plot. Explain how variable scaling, axis order, and sample size affect what you see.

Part D — Critical Review

Take one multivariate chart and redesign it as two coordinated simpler charts. Explain what became easier to interpret and what information was lost.

Coding Project — Multivariate Visualization Toolkit

Extend the cumulative EDA master notebook with a multivariate visualization section. Reuse your distribution and relationship plotting functions from Chapters 10 and 11.

- Create a bubble-scatter function with explicit size and color mappings and a clear legend.
- Create a faceting helper for grouped scatterplots or distribution plots with configurable shared axes.
- Create a heatmap helper that requires the aggregation or normalization rule to be explicit.
- Create a correlation heatmap with selectable correlation method and a note about pairwise missingness.
- Create a parallel-coordinate helper that allows an explicit variable list and optional display scaling.
- Add sample-size and missingness reporting to every multivariate workflow.
- Test the functions on a dataset with mixed variable types and at least four meaningful variables.
- Produce two coordinated views rather than relying on a single overloaded chart.

Expected reusable components: plot_bubble_relationship(), plot_faceted_relationship(), plot_summary_heatmap(), plot_correlation_heatmap(), and plot_parallel_coordinates(). Keep raw data unchanged and return figure or grid objects for customization.

Deliverables: updated cumulative EDA master notebook, reusable plotting functions, saved figures, and a short report explaining encodings, findings, and limitations.

Preview of Next Chapter

Chapter 13 — Correlation and Association Measures

We'll move from visual evidence to quantitative measures of association. Topics include covariance, Pearson correlation, Spearman and Kendall rank correlations, categorical association, effect-size interpretation, assumptions, outlier sensitivity, and the limits of correlation. We'll distinguish linear association, monotonic association, general dependence, and causation.